

Reg. No.

**B.Tech/ M.Tech (Integrated) DEGREE EXAMINATION, MAY 2024**  
Sixth Semester

**21CSC304J – COMPLIER DESIGN**

*(For the candidates admitted from the academic year 2022-2023 onwards)*

**Note:**

- (i) **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40<sup>th</sup> minute.
- (ii) **Part - B** and **Part - C** should be answered in answer booklet.

Time: 3 Hours

Max. Marks: 75

**PART – A (20 × 1 = 20Marks)**

Answer **ALL** Questions

- |                                                                                                                                                                                                                                                                                                                                     | Marks | BL | CO | PO |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----|----|----|
| 1. Which of the following is a cousin of the compiler?<br>(A) Interpreter (B) Debugger<br>(C) Text editor (D) Operating system                                                                                                                                                                                                      | 1     | 1  | 1  | 1  |
| 2. What is the purpose of the front-end of a compiler?<br>(A) To optimize the code (B) To generate intermediate code<br>(C) To perform analysis on the source code (D) To generate machine code                                                                                                                                     | 1     | 1  | 1  | 1  |
| 3. Which of the following regular expressions matches all the strings that starts with ab and ends with b over the input $\Sigma = \{a,b\}$<br>(A) $ab(a+b)b$ (B) $ab(a+b)^*b$<br>(C) $ab^*b$ (D) $ab^+b$                                                                                                                           | 1     | 1  | 1  | 2  |
| 4. Consider the following automata:<br><div style="text-align: center; margin: 10px 0;"> <pre> graph LR     start(( )) --&gt; q0((q0))     q0 -- a --&gt; q1((q1))     q1 -- a --&gt; q2(((q2)))             </pre> </div> It accepts<br>(A) Exactly two a's (B) All even number of a's<br>(C) Odd number of a's (D) At least one a | 1     | 1  | 1  | 2  |
| 5. What does error handling in parsing involve?<br>(A) Correcting syntax errors automatically (B) Ignoring syntax errors<br>(C) Detecting syntax errors and providing meaningful messages (D) Avoiding syntax error during error code execution                                                                                     | 1     | 1  | 2  | 1  |
| 6. Which of the following represents a valid production rule in a grammar?<br>(A) $E \rightarrow E + E * E$ (B) $E + E * E \rightarrow E$<br>(C) $E = E + E * E$ (D) $E \Rightarrow E$                                                                                                                                              | 1     | 1  | 2  | 2  |

7. Given the grammar:  
 $E \rightarrow E+T \mid T$   
 $T \rightarrow T * F \mid F$   
 $F \rightarrow (E) \mid id$   
 Which non-terminal should be recursive descent parser expand first when parsing the input  $id+id*id$ ?  
 (A) E (B) T  
 (C) F (D) id
8. Consider the following grammar  
 $S \rightarrow AB$   
 $A \rightarrow aB \mid \epsilon$   
 $B \rightarrow b$   
 What is follow(B)?  
 (A) {b} (B) {a}  
 (C) {\$} (D) {\$,a}
9. Which of the following parsing techniques employ bottom-up parsing?  
 (A) LL(1) (B) LR(0)  
 (C) Predictive parsing (D) Recursive descent parsing
10. What is handle pruning in shift-reduce parsing?  
 (A) Removing unnecessary symbols from the parse stack  
 (B) Trimming the input string to improve parsing efficiency  
 (C) Reducing the number of reductions performed during parsing  
 (D) Adjusting the grammar rules to reduce conflicts in parsing
11. In SLR parsing, what does 'S' in SLR stand for?  
 (A) Simple (B) Shift  
 (C) State (D) Syntax
12. What does LR(1) parsing use in addition to LR(0) parsing?  
 (A) Lookahead symbols (B) Reductions  
 (C) Production rules (D) Grammar symbols
13. Generate intermediate code representation for  $(a+b)*(c-d)$   
 (A)  $T1=a+b; T2 = c-d; T3 = T1*T2$  (B)  $T1 = a; T2 = +b; T3 = T1-T2;$   
 $T4 = c-d; T5 = T3*T4$   
 (C)  $T1 = (a+b)*; T2 = T1*(c-d)$  (D)  $T1 = a+; T2 = c-; T3 = b/d;$   
 $T4 = T1T3*T2T3$
14. Which notation is known as polish notations?  
 (A) Prefix (B) Infix  
 (C) Postfix (D) Reverse prefix
15. Convert the following infix to postfix  $a+b*c-d$   
 (A)  $abcd+*-$  (B)  $abc+*d-$   
 (C)  $+*-abcd$  (D)  $ab+*cd-$

16. What is the purpose of a register descriptor on code generation? 1 1 4 1  
 (A) To allocate memory for variable (B) To track the usage of register during code generation  
 (C) To optimize instruction scheduling (D) To manage runtime exceptions
17. Which of the following is the principal goal of code optimization? 1 1 5 1  
 (A) Increasing code readability (B) Reducing code size  
 (C) Increasing compilation time (D) Reducing program execution speed
18. What optimization can be applied on the following code? 1 1 5 1  
`mov a, 10`  
`add b, a`  
`mov a, 10`  
`add b, a`  
 (A) Loop unrolling (B) Constant propagation  
 (C) Common-sub expression elimination (D) Function inlining
19. How can this be optimized? 1 1 5 2  
`z = z * y;`  
`a = x;`  
`w = x * y + 6;`  
 (A) Remove `a = x` & change `w = z + b` (B) Remove `z = x * y`  
 (C) Change `w = x * y + 6` to `w = z + b` (D) It is already optimized only
20. Which type of graph is typically used to represent flow graph? 1 1 5 2  
 (A) Directed Acyclic Graph (DAG) (B) Direct graph  
 (C) Undirected graph (D) Bipartite graph

### PART - B (5 × 8 = 40 Marks)

Answer ALL Questions

21. a. What are the different phases of compiler? Explain the phases in detail. Write down the output of each phase for the expression:  $a = b + c * 50$ . 8 2 1 1
- (OR)
- b. Given the regular expression  $(a/b)^*abb(a/b)^*$ , construct its equivalent NFA. 8 2 1 1
22. a.i. Define CFG. 2 1 2 1
- ii. Find the first and follow of the following CFG. 6 2 2 2  
 $E \rightarrow E + T$   
 $E \rightarrow \bar{I}$   
 $T \rightarrow T * F$   
 $T \rightarrow F$   
 $F \rightarrow (E)$   
 $F \rightarrow id$

(OR)

b.i. What is an ambiguous grammar? Is the following grammar ambiguous?  $E \rightarrow E+E | E * E | (E) | id$  prove the answer. 6 2 2 2

ii. What is a parse tree? 2 1 2 1

23. a. What is an LALR(1) grammar? Construct LALR parsing table for the following grammar. 8 2 3 2

$S \rightarrow cC$

$C \rightarrow cC$

$C \rightarrow c/d$

(OR)

b. What is an LR(0) item? Construct an SLR parsing table for the grammar. 8 2 3 2

$G:$

$S \rightarrow L = R | R$

$L \rightarrow *R | id$

$R \rightarrow L$

24. a. What are different intermediate code forms? Discuss different types of implementations of three address statements. 8 2 4 1

(OR)

b. What do you mean by attribute grammars? Discuss the translation scheme for converting on infix expression to its equivalent postfix form. 8 2 4 1

25. a. Explain the following code-optimization techniques with examples 8 2 4 1

(i) Constant propagation

(ii) Strength reduction

(OR)

b.i. Discuss in detail about global dataflow analysis. 6 1 5 1

ii. Define Peep-hole optimization. 2 1 5 1

**PART – C (1 × 15 = 15 Marks)**

Answer ANY ONE Question

Marks BL CO PO

26. Give the expression  $y = (a+b)*c$  in 15 2 4 4

(i) Post fix notation

(ii) Abstract syntax tree

(iii) Three address code, implemented in triples

27. Write the syntax directed translation scheme with backpatching to generate three address code for the given grammar. 15 2 4 4

$S \rightarrow \text{while } E \text{ do } \hat{S} \text{ being } L \text{ end}$

$L \rightarrow L; S | S$

$E \rightarrow E \text{ or } E | E \text{ and } E | \text{not } E | id$

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